

Total No. of Questions : 4]

SEAT No. :

PD257

[Total No. of Pages : 2

[6411]-32

B.E. (Civil) (Insem)

HYDROPOWER ENGINEERING

(2019 Pattern) (Semester-VIII) (Elective-V) (401013 E)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn whenever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator (non-programmable) and steam tables is allowed.

Q1) a) Explain the present status of hydropower generation in India. **[5]**

b) What is surge tank? What are different types of surge tank? Explain with neat sketches. **[5]**

c) Write short note on participation of private sector in hydropower development. **[5]**

Q2) a) What is pump storage peak load plants? Explain with the help of neat sketch. **[5]**

b) What is greenhouse effect? Write short note on it. **[5]**

c) Draw a schematic layout of hydroelectric power plant. Explain the penstock. **[5]**

Q3) a) Define: **[5]**

- i) Plant capacity factor
- ii) Plant use factor
- iii) Connected load factor
- iv) Demand factor
- v) Diversity factor

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- b) Write the significance of diversity factor on the cost of hydropower generation. [5]
- c) When runoff river plant operates as peak load station with weekly load factor of 30%, all its capacity is firm capacity. What must be minimum flow in the river so that the station may serve as the base load station? It is given that rated installation capacity of generator is 10,000 KW. Operating head is 25m. Plant efficiency is 90%. Estimate the daily load factor of plant if the stream flow is 20 cumec. [5]

OR

- Q4) a) Write short note on tariff for electrical energy and types of tariffs. [5]
- b) Write note on prediction of load with suitable load curve diagram. [5]
- c) A load on hydroelectric plant varies from minimum 15,000 KW to a maximum 60,000 KW. Two turbo generators of capacities 30,000 KW each have been installed. Calculate: [5]
- Total installed capacity of plant
 - Plant factor
 - Maximum demand
 - Load Factor
 - Utilization factor